



NAVIGATING FORENSIC DNA EVIDENCE: GUIDELINES, CHALLENGES, AND SOLUTIONS

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ABSTRACT

Abstract This article outlines the importance of forensic DNA evidence in criminal investigations, including its types and characteristics. Additionally, through an analysis of the current practices of forensic DNA evidence in criminal proceedings, detailed discussions on its application rules are provided. This includes the extraction and submission of biological samples, standardization of examination and identification, as well as issues regarding courtroom scrutiny and authentication rules. Finally, the article proposes some improvement suggestions aimed at meeting the needs of criminal proceedings in our country and making beneficial explorations for the application rules of forensic DNA evidence.

Introduction

Abstract This article outlines the importance of forensic DNA evidence in criminal investigations, including its types and characteristics. Additionally, through an analysis of the current practices of forensic DNA evidence in criminal proceedings, detailed discussions on its application rules are provided. This includes the extraction and submission of biological samples, standardization of examination and identification, as well as issues regarding courtroom scrutiny and authentication rules. Finally, the article proposes some improvement suggestions aimed at meeting the needs of criminal proceedings in our country and making beneficial explorations for the application rules of forensic DNA evidence.

Methodology

This study employs a methodology of literature review and analysis, with the importance of forensic DNA evidence in criminal investigations as the entry point, to explore its application rules in criminal proceedings. Specific methods include:

Literature review method: In-depth review and analysis of existing literature, legal provisions, and case laws to elucidate the importance of forensic DNA evidence and its application rules in criminal proceedings.

Case analysis method: Through case analysis, a thorough examination of the application status and related issues of forensic DNA evidence in actual cases is conducted, as well as the impact on courtroom scrutiny and authentication rules.

Qualitative research method: In-depth discussion and analysis of potential errors, sample sources and handling, contamination and cross-contamination, professional opinion disputes, legal, and ethical issues in the application process of forensic DNA evidence, revealing practical problems and challenges in criminal proceedings.

Comparative method: Drawing on research findings on forensic DNA evidence from developed countries such as Europe and America, adopting a scientific attitude and rational understanding to provide beneficial experiences and insights for criminal proceedings in our country.

Results

The importance of forensic DNA evidence in criminal investigations is increasingly recognized, but it is accompanied by a series of challenges. Firstly, forensic DNA evidence may face challenges in error rates and reliability. Although forensic DNA technology theoretically has high accuracy, there is still a certain error rate in practical applications, which may be influenced by factors such as sample quality, experimental procedures, and data interpretation. Secondly, there are issues with the sources and handling of forensic DNA technology samples. For example, unclear sample sources or contamination or deterioration during collection, preservation, and transportation processes may affect the identification of evidence. Additionally, forensic DNA evidence may suffer from contamination and cross-contamination leading to inaccurate DNA analysis results, which may confuse the court with professional opinions, affecting judgment outcomes. Finally, forensic DNA evidence also faces legal and ethical issues, including challenges in protecting personal privacy and rights, avoiding misuse or improper use of DNA evidence, among others.

To address these issues, first, it is necessary to improve the rules for the extraction and submission of biological samples, establish clear procedures and standards to ensure that collected biological samples are of high quality and reliability, and specify submission procedures and standards to ensure the smooth progress of subsequent DNA testing and identification work. Second, the establishment of standardized rules for examination and identification is essential to ensure the accuracy and reliability of DNA identification results. This includes establishing standardized procedures, methods, and result confirmation for DNA examination and identification, as well as certifying and training DNA identification personnel to ensure the accuracy and reliability of examination and identification work. Additionally, the importance of establishing a sound courtroom scrutiny system and authentication rules should not be overlooked to ensure the legality and reliability of DNA evidence. This includes establishing strict procedures and standards for the scrutiny of DNA evidence and the authentication system for DNA evidence, as well as requirements for the testimony of DNA identification personnel in court. Through these solutions, it is hoped that the credibility and reliability of forensic DNA evidence can be improved to ensure judicial fairness and legal justice.

Discussion

Discussion The development of forensic DNA technology can be traced back to the early 1980s. One important breakthrough was in 1984 when British scientist Alec Jeffreys invented DNA fingerprinting technology, which is widely regarded as the cornerstone of modern forensic DNA analysis. As an extremely important type of evidence, forensic DNA evidence plays a crucial role in solving various complex cases. Through research on this type

of evidence in developed countries such as Europe and America, we still need to treat it with a scientific attitude and rational understanding, realizing the potential risks of scrutiny and quality certification. In view of this, this study delves into the issues of forensic DNA evidence application in criminal proceedings, with a particular focus on a thorough analysis of the application rules of forensic DNA evidence.

Forensic DNA evidence is obtained by professional identification personnel using forensic identification techniques to interpret the genetic information of DNA molecules left at the crime scene, thereby obtaining interpretive information to prove the facts of the case. Due to the relatively short development time of forensic DNA evidence, it still faces some problems in judicial practice, such as: Error rates and reliability: Although forensic DNA technology theoretically has high accuracy, there is still a certain error rate in practical applications. This may be influenced by factors such as sample quality, experimental procedures, and data interpretation. Therefore, caution is needed in explaining and using DNA evidence in court to avoid misleading court judgments. Sample sources and handling: The source and handling of DNA samples may affect their reliability and credibility. For example, if the sample source is unclear or if the sample is contaminated or deteriorates during collection, preservation, transportation, etc., it may lead to problems with DNA analysis results, thereby affecting the identification of evidence. Contamination and cross-contamination: In cases of multiple sample mixtures or cross-contamination during sampling and processing, DNA analysis results may be inaccurate. Therefore, strict measures need to be taken during sample collection and processing to avoid contamination and cross-contamination. Disputes over professional opinions: There may be different professional opinions on the interpretation and analysis of DNA evidence, and different identification personnel or laboratories may reach different conclusions. In court, conflicting or disputed expert opinions may confuse judges and juries, affecting judgment outcomes. Legal and ethical issues: When using forensic DNA technology, it is necessary to strictly comply with relevant laws, regulations, and ethical guidelines to protect personal privacy and rights, and to avoid misuse or improper use of DNA evidence. Additionally, the relationship between personal privacy and social public interests needs to be balanced to ensure the legal and ethical use of technology.

Therefore, we should improve the application rules of forensic DNA evidence:

Rules for the extraction and submission of biological samples: Biological sample extraction is the first step in the formation of forensic DNA evidence, and its impartiality directly affects the credibility and availability of DNA evidence. Therefore, in the rules for the extraction and submission of biological samples, we need to establish clear procedures and standards to ensure that the extraction and submission of biological samples in the case investigation process are strictly carried out according to regulations. Before conducting biological sample extraction, a comprehensive assessment of the case scene should be conducted to ensure that the collected biological samples are of high quality and reliability. At the same time, for the submission process, clear procedures and standards need to be specified to ensure the timeliness and accuracy of submissions, thereby ensuring the smooth progress of subsequent DNA testing and identification work.

Standardization rules for examination and identification: In order to ensure the accuracy and credibility of DNA identification results, it is necessary to establish standardized rules for examination and identification. This includes establishing standardized procedures,

methods, and standards for DNA examination and identification, as well as certifying and training DNA identification personnel to ensure their professional level and the scientific and objective nature of identification results. At the same time, it is also necessary to clarify the requirements for the construction and management system of DNA identification laboratories to ensure that the facilities and equipment of the laboratories comply with relevant standards, ensuring the accuracy and reliability of examination and identification work.

Courtroom scrutiny system and authentication rules: In the application process of DNA evidence, it is necessary to establish sound courtroom scrutiny systems and authentication rules to ensure the legality and reliability of DNA evidence. This includes establishing strict procedures and standards for the scrutiny of DNA evidence, ensuring the legal acquisition and use of evidence, as well as establishing a certification system for DNA evidence to ensure the authenticity and credibility of evidence. Additionally, the procedures and requirements for the testimony of DNA identification personnel in court need to be clarified to ensure the truthfulness and credibility of their testimonies, thereby ensuring the fairness and credibility of courtroom judgments.

Conclusion

While we highly commend the powerful evidential force of forensic DNA evidence, we must also recognize that improper use of forensic DNA evidence in judicial practice may lead to wrongful convictions. Therefore, we need to adopt a scientific and rational attitude towards forensic DNA evidence, recognize its limitations, and examine various problems existing in the application of forensic DNA evidence. In judicial practice, to ensure the probative force of forensic DNA evidence, we need to start from preventing wrongful convictions, and comprehensively improve the rules for the application of forensic DNA evidence in criminal proceedings based on relevant legal norms. This includes establishing standardized processes and standards from application to identification, and then to review and scrutiny, providing insights for preventing and correcting wrongful convictions caused by improper use of forensic DNA evidence. Through such efforts, we can better safeguard judicial fairness and legal justice.

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